

Oneiric Dynamics for Artificial Intelligence

A Research Framework for Offline Generative Recombination, Memory Consolidation, and Constrained Self-Revision

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Abstract

Internally generated experience is already central to generative replay, model-based planning, and several dream-inspired learning methods. What remains less settled is how to distinguish the functions of replay, recombination, and counterfactual generation, and how to evaluate their effects without allowing synthetic material to become its own evidence. This paper develops Oneiric Dynamics (OD) as a research framework for that question. It describes an offline phase in which a provenance-preserving mixture generator proposes candidate experiences; an absolute admissibility rule precedes any relative weighting; and proposed parameter and memory updates remain subject to explicit revision limits and external checks. The formulation distinguishes a penalty that discourages change from a condition that actually rejects a change. Elementary results establish mixture validity, conditional parameter-displacement bounds, and a memory-mixture bound. A standard invariant-measure result is stated only for implementations whose complete state and transition law satisfy the required assumptions. Analytic counterexamples identify limitations of support-based novelty, normalized scores, and soft anchoring. A comparative experimental program then asks whether controlled recombination provides benefits beyond replay and ordinary augmentation, in which task regimes, and at what computational cost. No empirical advantage is reported. The intended contribution is a common set of distinctions, testable mechanisms, and failure criteria with which a more cumulative investigation of internally generated learning might proceed.

Keywords: internally generated learning; artificial dreaming; generative replay; compositional generalization; memory consolidation; provenance; proximal regularization; Markov processes.

1. Introduction

A learning system can revise what it knows not only by encountering new observations, but also by revisiting and reorganizing material already available to it. Wake–sleep learning, generative replay, and model-based planning provide established examples of this broader design space [1,4,18]. The question is therefore not whether artificial systems can perform useful internal computation. It is which forms of internal generation contribute something beyond rehearsal, and under what conditions those contributions remain compatible with externally supported knowledge.

The term *dreaming* can obscure this question when it groups together exact replay, synthetic reconstruction, latent interpolation, imagined control trajectories, and speculative hypothesis generation. These operations need not have the same effects. Replaying a previous task can preserve a representation without teaching a new composition. A generated combination can expose a useful regularity while also introducing a false relationship. A model-relative alternative can be informative about an assumption without constituting evidence that the alternative occurred.

Oneiric Dynamics (OD) is proposed as a framework for separating these possibilities. It treats internal generation, admission for learning, state revision, and epistemic validation as distinct operations. The framework is broad at the level of its interfaces, but its mathematical guarantees are deliberately tied to specified implementations. A neural parameter update and a revision of a symbolic memory are not assumed to obey the same geometry merely because both can be described as learning.

The central research question is whether a controlled combination of replay, recombination, and counterfactual generation can improve retention or transfer at a cost that is justified relative to simpler alternatives. Answering it requires more than coherent notation. It requires operational definitions, competing baselines, externally grounded evaluation, and the possibility that an apparently productive internal phase should make no accepted change.

This article develops a specification for that investigation. It makes provenance part of the represented experience, provides an explicit reject-all outcome, separates soft regularization from acceptance constraints, and identifies the assumptions needed for local revision bounds and stochastic analysis. The elementary results characterize properties of these mechanisms using established tools from probability and optimization, clarifying which conclusions follow from the formulation and which remain empirical questions.

OD therefore frames a research space in which the value of internally generated learning can be tested through explicit comparisons, failure criteria, and reproducible experiments. Its usefulness will depend on whether these distinctions reveal effects that are not explained as well by simpler formulations.

2. Background and Relation to Prior Work

2.1 Biological inspiration without biological equivalence

Complementary learning systems theory links rapid acquisition of particular experiences with slower integration of structured knowledge [2,3]. Sleep research supplies evidence for memory reactivation and consolidation, but does not justify treating these processes as identical to the subjective experience of dreaming [9,13]. A meta-analysis reports an association between task-related dream content and subsequent memory performance [10]; methodological reviews emphasize the difficulty of interpreting such associations and of isolating the contribution of dream experience itself [9,11].

The overfitted-brain hypothesis proposes that altered dream statistics may support generalization [6]. Here it supplies a computational motivation for distinguishing rehearsal from perturbation, not a biological premise

on which the formal results depend. Likewise, the generative model of memory developed by Spens and Burgess relates reconstruction, imagination, and consolidation within a particular computational account [8]. These connections motivate questions about artificial systems without settling them.

Throughout this paper, *oneiric* names a functional computational regime. It does not imply consciousness, phenomenal dreaming, emotion, or an artificial counterpart of human sleep stages.

2.2 Replay, planning, and regularized learning

The wake–sleep algorithm alternates recognition- and generation-related updates [1]. Deep generative replay uses synthesized examples to support learning across task sequences [4], within the wider field of continual learning reviewed by Parisi and colleagues [5]. Sleep-like unsupervised replay has also been investigated experimentally as a means of recovering prior-task performance in neural networks [7]. Together, these lines of work establish internally generated training data as an important precursor to the questions addressed by OD.

Dyna integrates experience-based learning with planning through a learned model [18]. DreamerV3 similarly learns an environment model and improves behavior using imagined trajectories, together with mechanisms designed to stabilize learning [12]. Both ground their learned models in environmental experience. OD focuses on a complementary organizational question: how internal generation, provenance, candidate admission, and subsequent revision can be represented explicitly across implementations.

Constraining parameter change also has important antecedents. Elastic Weight Consolidation protects parameters according to their estimated importance for previous tasks [19]. Proximal methods formalize optimization relative to a reference point [20]. The isotropic quadratic penalty used below is a simple member of this established family and provides a transparent reference for studying local revision. Task-sensitive regularizers or behaviorally meaningful distances may be preferable in specific implementations.

2.3 Explicitly dream-inspired machine learning

Ni and Liu’s SleepNet and DreamNet study feature enrichment and reconstruction for visual classification [14]. NeuroDream [15] and Dreaming Machine Learning [16] provide additional dream-inspired conceptual antecedents.¹

Dream2Learn is a particularly relevant comparison. It generates synthetic auxiliary classes by steering a frozen generative model with a continual classifier, and reports improved performance and forward transfer in several evaluated classification settings [17]. Its mechanism already includes a stopping procedure and access to real experience. OD therefore differs most clearly in the proposed separation of generation modes, provenance, candidate admission, and state-revision rules across architectures. Dream2Learn’s reported results motivate comparison, while the mixture and admission rules introduced below remain to be evaluated independently.

2.4 Intended scope of the contribution

The individual ingredients of OD have substantial precedents. The proposed contribution lies in organizing them into a specification in which different sources of internal experience, acceptance conditions, revision mechanisms, and evidence statuses can be varied separately. The value of this organization is an empirical and methodological question to be tested against existing formulations.

¹ The cited version of [14] is arXiv:2409.01633v4, published in April 2026; its initial 2024 release used the title Dreaming is All You Need. References [15] and [16] are preprints and are included here as conceptual antecedents rather than empirical validation of OD.

Established line of work	Question emphasized by OD
Wake–sleep and generative replay [1,4]	When does a transformed internal experience contribute beyond reconstruction or rehearsal?
Dyna and world-model imagination [12,18]	Which generation, grounding, and revision mechanisms can be isolated for comparison across task settings?
Sleep-inspired consolidation [7,8,14]	Which effects depend on replay, recombination, selection, or the particular learning architecture?
Dream2Learn [17]	Can any benefit of novel synthetic compositions extend beyond its demonstrated classification mechanism?
Parameter regularization and proximal methods [19,20]	What does a revision bound control, and what still requires behavioral testing or rejection?

Table 1. Relationship of OD research questions to representative prior approaches.

3. Operational Definition of an Oneiric Phase

An OD phase is an identifiable interval in which a system proposes learning material from its existing model and memory, without requiring newly acquired environmental observations to generate that material. Previously acquired external observations and fixed checks may still be used for evaluation. Here *offline* describes the source of candidate experience, not necessarily network disconnection or execution during idle time.

Four conditions define the proposed regime. First, candidates arise from already available state rather than a new observational batch. Second, the available generation mechanism permits substantive transformation, such as recombination or a model-relative change of conditions, rather than exact recall alone. Third, the protocol distinguishes generating a candidate from admitting it for a particular learning use. Fourth, any accepted revision remains subject to declared reference checks and revision limits. A phase may validly end with no parameter or memory change.

Replay-only operation is retained as a limiting case and experimental baseline. The definition does not require every batch to contain a novel accepted item, and it does not claim a sharp ontological boundary between replay, reconstruction, and augmentation. Operator labels refer to how an item was produced; novelty is measured separately.

A candidate admitted for representation learning is not automatically a verified statement about the world. For example, a transformed image may be useful for training under a known label-preserving rule, without depicting an observed event. Similarly, a counterfactual trajectory can be valid under its declared intervention while differing from the recorded trajectory. Its evaluation must preserve that qualification.

4. Formal Framework

4.1 State, evidence, and cycle structure

Let $\Theta \subseteq \mathbb{R}^p$ be a parameter space and let $\mathcal{X}$ be an experience space. Experiences may be vectors, token sequences, trajectories, or structured objects, provided the required measurable structure is defined. To preserve origin information, use a marked space $\mathcal{R} = \mathcal{X} \times \mathcal{L}$, where $\mathcal{L}$ contains provenance and scope metadata. A record $x = (z, l)$ distinguishes its observational or synthetic origin, generating operation, source references, model version, and any intervention assumptions.

Origin and validation status are different attributes. Exact replay retains the provenance of the original record and does not count as a new independent observation. A reconstructed replay sample is synthetic even when its purpose is rehearsal. Independent checking may add evidence or restrict a claim’s scope; it does not change the historical origin of the record.

Write the cycle state as

$$S_k = (\theta_k, \mu_k, \mathcal{A}_k, h_k), \mu_k \in \mathcal{P}(\mathcal{X}). \quad (1)$$

Here $\mathcal{A}_k$ contains the externally grounded reference material and its versions. The auxiliary state h_k includes whatever additional information is necessary to determine the law of the next update, such as optimizer moments, scheduler state, generator state not contained in θ_k , and relevant environmental or controller variables. This is a requirement for state sufficiency, not a claim that the same finite state representation works for every architecture.

An external update produces the reference state for the internal phase:

$$S_k^W = W_k(S_k, D_k, \xi_k^W). \quad (2)$$

The batch D_k is newly available external experience; ξ_k^W denotes update randomness. Its conditional law and any temporal dependencies must be specified. The superscript W denotes the post-external-update reference, not a biological waking state. During the subsequent OD phase, reference data, admission criteria, provenance definitions, and the baseline used for comparison are frozen.

For evaluation, distinguish reference data used in fitting, $\mathcal{A}_k^{fit}$, from data or checks used in admission, $\mathcal{A}_k^{check}$. A final experimental test set is separate from both. These are logical roles; sharing a physical storage system does not justify sharing evaluation information.

4.2 A provenance-preserving mixture generator

Let K_{rep}, K_{rec}, K_{cf} be Markov kernels from the post-update state space to $\mathcal{X}$. Replay reconstructs or resamples prior experience. Recombination assembles components not previously observed in that particular conjunction. Counterfactual generation changes a declared condition, action, or modeling assumption to produce a model-relative alternative.

The last term is deliberately restricted. A generated alternative is not an identified causal counterfactual merely because it is labeled as one. Any claim about interventions in the actual world requires additional causal assumptions and evidence. Without them, the kernel supplies hypothetical scenarios for investigation.

For nonnegative coefficients summing to one, define

$$Q_k(B) = \alpha_k K_{rep}(S_k^W, B) + \beta_k K_{rec}(S_k^W, B) + \gamma_k K_{cf}(S_k^W, B). \quad (3)$$

$$\alpha_k + \beta_k + \gamma_k = 1, \rho_k = \beta_k + \gamma_k. \quad (4)$$

The coefficients may be fixed or measurable functions of the sufficient state. For the reference construction, draw a finite batch $x_1, \dots, x_m$ conditionally independently from Q_k . The number of candidates and any further attempts are limited by a declared budget. Overlapping output distributions are allowed: the labels identify generation procedures, not mutually exclusive sets of experiences.

4.3 Absolute admission before relative weighting

Let $a_k(x) \in \{0, 1\}$ be a measurable candidate-admission rule evaluated against the frozen reference. It checks the intended learning use, required structure, provenance, applicable constraints, and sufficiently established compatibility with external anchors. The conservative reference policy does not admit a candidate whose required checks remain unresolved. Such a candidate may remain an explicitly provisional research artifact, but it supplies no learning weight in the current phase.

Define the admitted index set

$$\mathcal{J}_k = \{j \in \{1, \dots, m\} : a_k(x_j) = 1\}. \quad (5)$$

If $\mathcal{J}_k$ is empty, the learning update is skipped. This case is essential: a normalized ranking cannot determine whether an entire batch should be rejected.

For admitted candidates, let N_k denote a novelty score, C_k a coherence cost, and A_k a cost of disagreement with applicable external checks. One optional ranking is

$$r_k(x) = \lambda_N N_k(x) - \lambda_C C_k(x) - \lambda_A A_k(x), \quad (6)$$

with finite scores and nonnegative weights calibrated on development data. When $\mathcal{J}_k \neq \emptyset$, a numerically stabilized Gibbs weighting is

$$\omega_{k,j} = \frac{\exp((r_k(x_j) - r_k^{\max})/T)}{\sum_{i \in \mathcal{J}_k} \exp((r_k(x_i) - r_k^{\max})/T)}, j \in \mathcal{J}_k, T > 0. \quad (7)$$

Here $r_k^{\max} = \max_{i \in \mathcal{J}_k} r_k(x_i)$, and weights outside $\mathcal{J}_k$ are zero. Invalid or nonfinite scores are handled as explicit evaluation failures. Uniform weighting over admitted candidates is a simpler alternative and should be included in experiments.

The admission rule supplies absolute restrictions; the weights express relative preference among survivors. A large novelty score cannot compensate for a failed mandatory check. For counterfactual records, disagreement is evaluated against constraints that remain applicable under the declared intervention, rather than treating every departure from the observed event as an error. The weighted empirical distribution is

$$\hat{Q}_k = \sum_{j \in \mathcal{J}_k} \omega_{k,j} \delta_{x_j}. \quad (8)$$

This is a distribution of admitted learning material, not a distribution of newly observed facts.

4.4 Parameter proposals: regularization and explicit limits

For an admitted batch, let $\ell_O(\theta; x)$ be its task-specific learning loss and let $R_A(\theta; \mathcal{A}_k^{fit})$ be a loss on fitting anchors. Freeze candidate weights during the proposed parameter update and define

$$f_k(\theta) = \sum_{j \in \mathcal{J}_k} \omega_{k,j} \ell_O(\theta; x_j) + \lambda_G R_A(\theta; \mathcal{A}_k^{fit}), \lambda_G \geq 0. \quad (9)$$

A proximal objective discourages displacement from the reference parameters:

$$J_k(\theta) = f_k(\theta) + \frac{\lambda_p}{2} \|\theta - \theta_k^w\|_2^2, \lambda_p > 0. \quad (10)$$

The idealized proposal minimizes J_k over Θ . A practical optimizer may instead return θ_k^* satisfying the checkable reference-value condition

$$J_k(\theta_k^*) \leq J_k(\theta_k^W) + \varepsilon_k, \varepsilon_k \geq 0. \quad (11)$$

This condition is weaker than certifying a global optimum. When objective values are estimated, their uncertainty must be accounted for before claiming that the condition has been verified. A separately specified radius condition, $\|\theta_k^* - \theta_k^W\|_2 \leq r_{\theta,k}$, can directly reject excessive movement, whether or not the optimization method provides a useful theoretical certificate.

The anchor penalty and the proximal penalty have different roles. The first encourages compatibility with fitting evidence; the second discourages parameter displacement. Neither one, with a finite coefficient, guarantees semantic correctness or non-regression. Acceptance of the full proposed state is a further operation, defined below.

4.5 Memory proposals and preservation of source status

A general consolidation operator can preserve, merge, compress, reweight, or delete records. Without constraints on that operator, a parameter bound says nothing about memory revision. For an analytically transparent reference case, restrict consolidation to a mixture on the same marked space:

$$\mu_k^* = (1 - \eta_k) \mu_k^W + \eta_k \hat{Q}_k, 0 \leq \eta_k \leq \bar{\eta}_k \leq 1. \quad (12)$$

This updates a learning distribution without relabeling the origin of its records. Protected external source material remains available independently of its current sampling weight. When the admitted set is empty, define $\mu_k^* = \mu_k^W$ and do not construct an empty probability distribution $\hat{Q}_k$.

Equation (12) is an example, not a universal memory architecture. A finite buffer, a changing retrieval index, or a generative memory must be analyzed through its own update rule. In particular, replacing records with compressed approximations is not covered by the mixture bound unless the compression error is itself controlled in the stated metric.

4.6 Joint acceptance and rollback

Let $u = (\theta, \mu)$ denote the learning state. The proposed pair u_k^* is compared with u_k^W , the post-update reference. For each declared protected check q , let $L_q(u; \mathcal{A}_k^{check})$ be a loss, with lower values preferred, and let $\delta_{q,k} \geq 0$ be a prespecified tolerance. A measurable gate g_k accepts only when all required conditions hold, including

$$L_q(u_k^*; \mathcal{A}_k^{check}) \leq L_q(u_k^W; \mathcal{A}_k^{check}) + \delta_{q,k}. \quad (13)$$

Applicable structural invariants, provenance requirements, and revision radii are checked as well. Missing evidence, an unresolved required result, or a violated condition gives $g_k = 0$. The learning-state transition is

$$g_k = 1 \Rightarrow u_{k+1} = u_k^*, g_k = 0 \Rightarrow u_{k+1} = u_k^W. \quad (14)$$

The symbol u_{k+1} in (14) denotes the pair $(\theta_{k+1}, \mu_{k+1})$. The external anchors remain unchanged during this internal phase. Auxiliary records can record cost, rejection, and diagnostics according to the declared protocol; a no-change learning decision does not mean that no computation occurred.

A gate establishes compatibility with the checks actually applied. It is not a proof of correctness on unobserved tasks. Repeated adaptive use of admission data can also overfit those data, which is why a sealed final test set remains necessary. Thresholds, reference data, and evaluator definitions cannot be revised by the candidate merely to obtain acceptance.

4.7 When a Markov description is justified

If the stored state is sufficient and the generation, optimization, consolidation, and acceptance procedures define measurable transitions, the complete cycle induces a kernel

$$S_{k+1} \sim P_k(S_k, \cdot). \quad (15)$$

Randomness alone does not establish this property. Optimizer history or temporal environmental dependence omitted from S_k can make the represented process non-Markovian. Likewise, writing an optimizer as an argmin does not supply a measurable selection when minima are nonunique; an implementation must specify its selection rule or its stochastic update law.

A time-homogeneous kernel $P_k = P$ requires compatible time-homogeneous environmental and update laws on the chosen state. Augmenting the state can represent some history dependence, but does not automatically make the state compact or create an invariant distribution. With changing tasks, expanding provenance histories, drifting data, or evolving generators, nonstationary analysis may be the appropriate starting point.

5. Conditional Mathematical Properties

The following results establish limited properties of the specified constructions. They do not imply the empirical hypotheses in Section 7.

5.1 Proposition 1: validity of the mixture kernel

Statement. Suppose the three component kernels in (3) are Markov kernels on the declared measurable spaces. If their weights are nonnegative measurable functions of the input state and sum to one, their weighted mixture is also a Markov kernel.

Proof. For a fixed state, a convex combination of probability measures is a probability measure. For a fixed measurable output set, each component probability is measurable in the input state; multiplication by measurable weights and finite summation preserve measurability. Both defining properties follow.

Interpretation. This is a validity result for the distributional specification. It does not establish validity or usefulness of any generated content.

5.2 Proposition 2: existence and a reference-value displacement bound

Statement. Let Θ be a nonempty closed subset of $\mathbb{R}^p$, with $\theta_k^w \in \Theta$. Suppose f_k is lower semicontinuous, finite at θ_k^w , and bounded below by a finite constant b_k on Θ . Then the objective (10) attains a minimum. Moreover, every proposal $\theta_k^* \in \Theta$ satisfying (11) obeys

$$\|\theta_k^* - \theta_k^w\|_2 \leq \sqrt{\frac{2(f_k(\theta_k^w) - b_k + \varepsilon_k)}{\lambda_p}}. \quad (16)$$

Proof. Since $f_k \geq b_k$, the quadratic term makes every finite sublevel set of J_k bounded. Lower semicontinuity and closedness of Θ make these sublevel sets closed, hence compact in finite dimensions. A nonempty finite sublevel set exists because the reference point has finite objective value, so a minimum is attained. For any proposal satisfying (11),

$$\frac{\lambda_p}{2} \|\theta_k^* - \theta_k^w\|_2^2 \leq f_k(\theta_k^w) - f_k(\theta_k^*) + \varepsilon_k \leq f_k(\theta_k^w) - b_k + \varepsilon_k,$$

which yields (16).

Scope. Convexity and an interior minimizer are not required for this statement. If f_k is convex and Θ is convex, the positive quadratic term also yields uniqueness. In a nonconvex problem, the existence of a global minimizer does not show that a practical solver finds one. The reference-value condition is a possible certificate for a returned proposal, not a convergence theorem for the solver. Proximal regularization itself is established methodology [20].

A separate, conditional gradient estimate remains useful. For an unconstrained or interior proposal with differentiable f_k , suppose the stationarity residual satisfies

$$\|\nabla f_k(\theta_k^*) + \lambda_p(\theta_k^* - \theta_k^w)\|_2 \leq \varepsilon_k^{stat}.$$

If $\|\nabla f_k(\theta_k^*)\|_2 \leq G_k$, the triangle inequality gives

$$\|\theta_k^* - \theta_k^w\|_2 \leq \frac{G_k + \varepsilon_k^{stat}}{\lambda_p}. \quad (17)$$

An exact stationary point has zero residual. An ex ante interpretation requires a suitable gradient bound established independently of the unknown proposal; otherwise (17) is mainly a conditional or retrospective statement. At a constrained boundary, this interior stationarity equation need not hold, whereas (16) still applies under its assumptions.

Neither estimate bounds semantic error or lifetime drift. Small accepted movements can accumulate, external-update steps are not bounded by these formulas, and a small Euclidean displacement can have a large effect on a sensitive decision boundary.

5.3 Proposition 3: memory-mixture displacement

Define $d_{TV}(\mu, \nu)$ as the supremum of $|\mu(B) - \nu(B)|$ over all measurable sets B , so that distances between probability measures lie in $[0, 1]$.

Statement. For the update in (12),

$$d_{TV}(\mu_k^*, \mu_k^w) = \eta_k d_{TV}(\hat{Q}_k, \mu_k^w) \leq \eta_k. \quad (18)$$

Proof. For every measurable B , the signed difference is $\mu_k^*(B) - \mu_k^w(B) = \eta_k[\hat{Q}_k(B) - \mu_k^w(B)]$. Take absolute values and the supremum over B .

This controls one update of the sampling distribution, not the integrity of every rare record or the accuracy of a learner trained from it. Under an additional compression step producing $\tilde{\mu}_k$ with $d_{TV}(\tilde{\mu}_k, \mu_k^*) \leq \kappa_k$, the triangle inequality gives a bound of $\eta_k + \kappa_k$. That premise can be restrictive: moving atoms in an empirical distribution may produce maximal total variation even when content changes little. A weaker transport or task-specific metric may be preferable, but requires its own assumptions and analysis.

5.4 Proposition 4: conditional existence of an invariant measure

Statement. Let the sufficient state space $\mathcal{S}$ be compact and metrizable, and let the time-homogeneous kernel P be Feller: $P\varphi$ is continuous whenever φ is continuous, where $P\varphi(s) = \int \varphi(t)P(s, dt)$. Then there is at least one probability measure π such that $\pi P = \pi$. This is the compact-state case of the Krylov–Bogoliubov result [21].

Proof. Fix $s \in \mathcal{S}$ and form the averaged transition laws

$$v_n = \frac{1}{n} \sum_{j=0}^{n-1} P^j(s, \cdot).$$

Compactness gives a weakly convergent subsequence with limit π . For continuous φ ,

$$|v_n(P\varphi) - v_n(\varphi)| = \frac{|P^n\varphi(s) - \varphi(s)|}{n} \leq \frac{2 \|\varphi\|_\infty}{n}.$$

The Feller property permits passage to the subsequential limit, yielding $\pi(P\varphi) = \pi(\varphi)$ and hence invariance.

Applicability. The theorem does not establish compactness or Feller continuity for OD as a whole. Threshold gates can introduce discontinuities, although discontinuous components do not by themselves prove that an integrated kernel fails to be Feller. Expanding record spaces, unbounded parameters, and auxiliary histories also require separate treatment. A finite-state instantiation or a suitably continuous compact-state model can satisfy the assumptions; applicability therefore requires verifying these conditions for the chosen implementation.

Existence does not imply uniqueness, convergence from every initial state, or a desirable regime. A reject-all identity transition, for example, admits every probability measure as invariant while performing no learning. If a complete cycle is deterministic and is additionally a contraction on a complete metric space, Banach’s theorem gives an attracting fixed point. A contraction property would need to be established separately for any OD implementation to which that conclusion is applied.

6. Novelty and the Interpretation of Control Parameters

6.1 Novelty relative to an explicit reference

Distance to the mathematical support of a memory distribution is not a generally useful novelty measure. A nondegenerate Gaussian has full support on its Euclidean space, so that distance is zero even for an extremely atypical point. OD therefore requires an implementation-specific reference rather than treating support distance as universal.

For a concrete option, let $B_k = \{z_1, \dots, z_n\}$ be a nonempty finite reference set selected from externally grounded experience, and let ϕ_k be an evaluation representation frozen during the phase. With a positive scale σ_k and a finite positive cap $N_{\max}$, define

$$N_k(x) = \min \left\{ N_{\max}, \frac{\min_{z_i \in B_k} \|\phi_k(x) - \phi_k(z_i)\|_2}{\sigma_k} \right\}. \quad (19)$$

The reference-set policy, representation, scaling, and cap must be specified before final evaluation. If the reference is empty or the scale cannot be defined, the measure is unavailable; setting it silently to zero would

confuse missing information with familiarity. Comparisons across phases require a common evaluation representation or an explicit recalibration procedure.

Equation (19) measures distance from sampled references, not novelty relative to an entire unknown population. Sparse coverage, nuisance features, and representation choice can distort it. On a factorized benchmark, a complementary measure can directly identify an unseen combination of individually known factors. For a density model, calibrated surprise relative to a declared typicality threshold is another option. These measures need not agree, and none is itself a measure of downstream utility.

6.2 Report proposal, admission, and influence separately

The exploratory share ρ_k in (4) records how frequently exploratory operators are selected, not how novel their outputs actually are. Report the realized operator counts and the batch admission fraction

$$\hat{p}_{adm,k} = |\mathcal{J}_k|/m,$$

alongside proposal novelty and weighted learning novelty:

$$I_k^{gen} = \frac{1}{m} \sum_{j=1}^m N_k(x_j), I_k^{learn} = \sum_{j \in \mathcal{J}_k} \omega_{k,j} N_k(x_j). \quad (20)$$

The second quantity is reported only when there is an admitted batch; otherwise its status is not applicable. The first already reflects the mixture used to sample candidates, so no further multiplication by ρ_k is built into its definition. Reporting the quantities separately prevents operator frequency, rejection, and learning weight from being confused.

Temperature affects relative weighting, not permission to violate a constraint. The mixture coefficients affect proposals, while acceptance tolerances and radii control which proposals are committed. Their effects should be investigated separately. Uniform weighting and simple gating provide reference comparators for testing whether a more elaborate scoring rule adds value.

7. Falsifiable Hypotheses and Experimental Program

The framework supports several research hypotheses; the mathematical results above do not entail them. Each hypothesis needs a specified task family, meaningful effect threshold, and evaluation budget. The scope supported by the framework should be updated in light of positive, null, or adverse findings.

7.1 Five hypotheses

H1 — Retention and compositional transfer may favor different mixtures. In task families that reuse known factors in new combinations, moderate recombination may improve transfer relative to matched replay, while replay may remain competitive for retention. Compare both outcomes on independent tasks. Any observed advantage should also be compared against ordinary augmentation and stronger replay baselines.

H2 — **A nonmonotonic novelty regime may exist.** Moderate realized novelty may be more useful than either near-identity rehearsal or highly distorted generation. This is a candidate inverted-U pattern, not a predicted universal law. Compare flat, monotone, and nonmonotone response models, and require any estimated interior optimum to lie within the tested range with appropriate uncertainty. The absence of this pattern is an informative result.

H3 — Proposal regularization and acceptance checks may have distinguishable effects. Removing anchor fitting or proximal regularization may increase adverse movement in proposed states; an unchanged hard gate

may convert that effect into more rejections rather than worse committed states. The analysis should therefore report both proposal-level movement and committed-state movement.

H4 — Transfer may depend on structural reuse. Benefits may be greater for new combinations of represented factors than for genuinely absent factors. The distinction must include the knowledge available in any pretrained generator: a factor unseen by the learner but already represented by its generator is not absent from the complete system.

H5 — Targeted internal generation may be more compute-efficient. Sampling around specified uncertainty or interference may yield more useful updates than uniform generation. Selection, scoring, verification, rejection, and storage costs must all be included. If targeting merely adds overhead or concentrates on unresolvable cases, this hypothesis fails in the evaluated regime.

7.2 Comparative conditions and ablations

Begin with a task family small enough to isolate these mechanisms. Use paired initializations and task sequences, repeated runs, matched access to external data, and comparable model capacity. The minimum comparisons are: no additional offline update; an equal-compute continuation or ordinary augmentation control; replay alone; replay with recombination; replay with counterfactual generation; and the full mixture. Uniform weighting over admissible candidates provides a direct comparator for score-based weighting.

Run targeted ablations of the fitting-anchor term, proximal term, candidate-admission rule, and joint commit gate in a controlled experimental environment. Record which other controls remain active. In particular, retaining a hard movement cap while removing a proximal penalty can conceal any difference in final displacement; retaining a rejection gate can conceal proposal degradation. The ablation report should show proposals, accepted states, rejection rates, and useful learning, not only the last accepted metric.

Generator architecture, pretraining access, memory capacity, data budget, and optimization effort must not change silently between conditions. Where Dream2Learn or another published method is technically comparable, implement its published procedure as faithfully as possible under matched resource and data conditions [17]. When faithful matching is not possible, document the mismatch and restrict the comparison accordingly.

Two cost views are useful: fixed total compute, which includes internal-generation overhead, and performance as a function of total compute. Matching only learner gradient steps can favor a method whose generator and verifier perform substantial additional work. Report unused budget and rejection costs rather than assuming every permitted internal phase must be filled.

7.3 A diagnostic factorized benchmark

One possible test uses a controlled renderer with independently specified factors such as shape, color, and position. External training contains only a subset of valid factor combinations while exposing each known factor value individually. Replay samples the observed combinations. Recombination constructs previously withheld combinations of known factors. Model-relative counterfactuals alter a declared factor or nuisance condition, with labels assigned by the benchmark’s explicit rule.

This design separates retention on seen combinations, generalization to held-out combinations of known factors, and adaptation when genuinely new factors arrive. It also makes ordinary compositional augmentation a strong baseline. The renderer is prior knowledge supplied to the experiment; its capabilities and labels must be available on the same terms to comparable methods. Any observed benefit would therefore concern the learning protocol rather than discovery of factors hidden inside the renderer.

Correctness comes from the declared generative task rule and an independent evaluator, not from the proposing model’s confidence. The final test instances are withheld from generation, tuning, admission, and early stopping. At least one complementary benchmark with learned or imperfect generative structure is needed before drawing conclusions beyond this controlled setting.

7.4 Outcome measures and separation of evidence

Primary outcomes should reflect the target task: retained-task loss or accuracy, compositional generalization, forward transfer when defined, and calibration on external test data. Define forward transfer relative to an appropriate matched learner before training on each incoming task, rather than reporting only accuracy after adaptation. Report mean effects, run-to-run dispersion, uncertainty intervals, and failure frequencies across paired task orders and seeds.

Process measures include admission rate, rejection reasons, proposal-to-commit differences, parameter displacement, memory displacement under the applicable metric, source-provenance integrity, and computational cost. Novelty is a diagnostic variable, not a success criterion. A system that rejects everything may pass all retention checks while contributing nothing; a system that accepts everything may improve its internal loss while degrading external performance.

Use development data for selecting hyperparameters, distinct admission data for operational checks, and untouched final data for the reported evaluation. Repeated access to admission data is an adaptive interaction and should be recorded. Task checks should include rare or protected cases where relevant, since an average loss can conceal a material regression. Report subgroup or case-level results when those requirements form part of the declared task.

7.5 Reproducibility and conditions for rejection

A future empirical study should release the generator definitions or code, data splits, task ordering, seeds, candidate and gate policies, optimizer settings, provenance schema, computational accounting, and analysis procedure. Thresholds and principal hypotheses should be specified before inspecting final test outcomes. Where preregistration is used, the dated registration should identify the hypotheses, primary outcomes, and analysis plan.

Evidence against a proposed OD instantiation would include no benefit over matched replay or augmentation, gains that depend on hidden access to future factors, unacceptable rejection or verification costs, degradation of protected outcomes, provenance loss, or benefits confined to the data used for selecting candidates. Such results would narrow or reject the tested mechanism; they would not by themselves decide every possible form of internally generated learning.

8. Failure Modes, Safety Constraints, and Epistemic Provenance

8.1 Self-generated evidence and correlated verification

The central epistemic risk is that a system both produces and endorses its own learning material. Multiple model instances do not necessarily provide independent evidence when they share data, architecture, prompts, or evaluative blind spots. Verification should therefore use external observations, independently specified task rules, executable checks, or appropriately independent review whenever those sources are available. Where they are not, the remaining uncertainty must be represented rather than converted into a factual label.

Agreement with known evidence and truth about an unobserved event are different claims. A coherent synthetic relation may remain unsupported. A useful training example can remain synthetic. The provenance record must preserve these distinctions through retrieval, replay, compression, and later reuse.

8.2 Recursive synthetic-data dependence

Research on recursively generated training data identifies regimes in which information about the original distribution is progressively lost [22]. This motivates explicit measurement of synthetic influence, source coverage, and diversity; it does not establish that every synthetic-data process collapses or that an anchor term prevents collapse.

Even the mixture in (12) can eventually replace most of a sampling distribution if it is repeated without refresh. With constant η and no compensating external update, the inherited weight of the original distribution after n mixture steps is $(1 - \eta)^n$. A small one-step bound therefore provides no permanent lower bound on original-data influence. Protected source retention, sampling requirements, and periodic external validation are separate design choices.

8.3 Counterfactual scope and invalid alternatives

A counterfactual must state what changes and what remains fixed. Changing an input while silently preserving an invalid label can manufacture a spurious learning signal. Conversely, rejecting every departure from a historical observation would defeat legitimate hypothetical exploration. Admission should evaluate the declared alternative within its permitted assumptions, while any resulting factual claim is separately checked against the external world.

When causal structure is uncertain, alternative models can be compared as hypotheses. Their outputs cannot be treated as measurements that resolve the causal uncertainty that generated them.

8.4 Operational boundaries and privacy

Internally generated work should not acquire new permissions merely because it is exploratory. For tool-using systems, authority, resource limits, external actions, and promotion of reusable knowledge require controls outside the generator’s own narrative. A hypothetical action is not authorization to execute that action. The parameter bounds in Section 5 establish no sandboxing or access-control guarantee.

Generated material can also reproduce sensitive source content. Evaluation should include checks appropriate to the data’s confidentiality and the intended release. Provenance may itself reveal sensitive associations, so preserving auditability does not imply disclosing all provenance to every consumer. Neither novelty nor a synthetic label establishes privacy protection.

8.5 Evaluation failures and false reassurance

Counterexamples in Appendix C show why three shortcuts fail: a support-distance score can miss atypicality; normalized weighting can fully endorse a bad batch; and finite penalties can permit regression. Additional failures arise when reference checks are obsolete, evaluator definitions drift, or rejected proposals are later reused as established facts.

These are not resolved by increasing the number of generated candidates. More internal computation can amplify an incorrectly specified objective. Stopping, requesting external evidence, or preserving the reference state must remain admissible outcomes.

9. Discussion: Framing a Research Space

OD invites investigation of a separation that can otherwise be lost inside an implementation: how experience is generated, why it is admitted, what state it may change, and what evidence supports retaining the change. The practical value of the framework can therefore be assessed by whether these distinctions improve explanation, experimental design, or observed outcomes relative to simpler descriptions.

This perspective treats self-improvement as a question of what internal computation can legitimately change. An internal phase may reveal consequences of existing knowledge, propose a new composition, or make a latent regularity easier to use. Such gains can arise from better organization or computation over existing information, while claims about new external facts still require suitable evidence.

For neural systems with trainable parameters, the proposed objective and local bounds are directly expressible. For systems that access a fixed model through an interface, changes may instead concern retrieval policies, prompts, workflows, or structured memories. Those are related applications of the proposal-check-revision distinction, but they do not inherit the Euclidean parameter theorem by analogy. Their state, admissible transitions, and evaluation criteria must be defined separately.

The most informative development path may therefore start small. A controlled experiment can establish whether recombination contributes beyond replay for one task family. Subsequent work can investigate imperfect anchors, selective generation, longer horizons, and changing environments. Evidence of benefits in one regime should guide, rather than substitute for, the next comparison.

10. Limitations and Scope

Empirical status. This manuscript presents definitions, conditional mathematical results, analytic examples, and an experimental program. Empirical evaluation of OD remains future work; the hypotheses about transfer, novelty, and efficiency are therefore open questions.

Theoretical scope. The mixture and displacement results characterize local properties of specified constructions. Practical convergence, useful long-run behavior, global robustness, alignment, and retention on unseen tasks require additional assumptions or empirical evidence. A Markov representation requires sufficient state; invariant-measure analysis further depends on conditions that must be established for each implementation.

Measurement and anchoring. Novelty depends on a representation, a reference, and a scale. Coherence and anchor disagreement require domain-specific implementations. A poor choice can make a mathematically valid rule scientifically uninformative. External anchors are also incomplete and potentially mistaken; satisfying them does not establish universal correctness. Repeated checking can overfit the admission set.

Memory and cumulative effects. The memory bound applies to the explicit mixture on a fixed marked space, not to unrestricted compression or deletion. One-step parameter and memory limits do not bound the entire future trajectory or every auxiliary state variable. Persistent provenance and immutable source records require an implementation capable of enforcing them.

Originality and generality. Internal generation, replay, planning, and regularization are established areas. OD’s proposed contribution is the organization of these mechanisms into a common specification and the experiments that distinguish their effects. The breadth and originality of that contribution should be assessed relative to the relevant adjacent literatures as the research program develops.

Biological interpretation. The analogy to dreaming is functional and partial. Nothing in the definition implies subjective experience or reproduces the causal role of dreaming in biological memory.

11. Conclusion

Artificial dreaming becomes a tractable research question when its mechanisms and claims are separated. Replay, recombination, and model-relative alternatives may serve different learning functions; generation does not establish admissibility; admissibility for learning does not establish factual truth; and a penalty that discourages revision does not replace a check that can reject it.

Oneiric Dynamics offers a formulation of these distinctions. Its mathematical results delimit several local properties, while its counterexamples identify guarantees that cannot be inferred from those properties. The resulting experimental questions concern where internal recombination adds value, how much external checking it requires, and when the simplest sufficient approach remains replay or no additional update.

The framework’s contribution will ultimately be measured by the research it makes possible: reproducible comparisons, better-characterized failures, and a clearer account of when internally generated experience helps rather than misleads. Positive, null, and adverse findings are all capable of moving that inquiry forward.

Research Materials and Disclosure

No empirical dataset, benchmark result, or tested OD software implementation accompanies this manuscript. The analytic examples are specified in Appendix C, and Appendix A defines the proposed reference procedure. Generative AI tools assisted manuscript drafting and revision. Responsibility for the scientific claims, equations, source selection, and interpretation remains with the author.

Appendix A. Reference Procedure

Algorithm A1 — One external-update/internal-generation cycle. Reference procedure for implementation and testing.

Inputs. A sufficient state S_k ; external batch D_k ; component kernels; mixture and weighting policies; fitting and checking anchors; finite compute budget; revision limits; and prespecified acceptance criteria.

1. Apply the external update to obtain S_k^w . Freeze the learning-state reference, source versions, evaluators, thresholds, novelty representation, and applicable provenance rules for this internal phase.
2. Confirm that internal generation addresses the declared task and that the remaining budget includes candidate evaluation and final checks. Otherwise retain the reference learning state.
3. Construct the mixture in (3). Draw at most the permitted number of marked candidates, recording operator, origin, source lineage, and hypothetical assumptions where relevant.
4. Apply absolute candidate admission. Retain unresolved candidates only as provisional artifacts outside the current learning update. If no candidate is admitted, return the reference learning state with a recorded no-change outcome.
5. Weight the admitted candidates uniformly or by (7), and construct (8). Do not permit relative weighting to override a failed mandatory check.
6. Propose parameters using the specified optimizer for (10). Evaluate the reference-value condition or the applicable optimization certificate and direct movement limits. Treat unavailable required certificates as a failure to admit the proposal, not as evidence that the bound holds.
7. Propose memory by (12), or by a separately specified and evaluated consolidation mechanism. Preserve source origin and any protected external records. Include compression or deletion effects when measuring revision.
8. Evaluate the joint proposed learning state against all required checks. Account for uncertainty in estimated losses and preserve evaluator independence from the candidate’s own justification.
9. Commit the learning-state pair only if the full gate passes. Otherwise restore the reference pair and discard provisional optimizer changes. Record diagnostics without silently changing the frozen acceptance policy.
10. Update auxiliary state according to its declared transition rule, retain the evidence for the decision, and end the phase. Further attempts require remaining budget and an explicitly permitted continuation.

Outputs. The next sufficient state, an accepted or no-change learning outcome, provenance-preserving records, check results, and total resource use. The returned state must include any auxiliary history that will affect future transitions.

Appendix B. Core Symbols

Symbol	Meaning
k	Index of the complete external-update/internal-generation cycle.
$\theta_k \in \Theta$	Trainable parameter state in the stated finite-dimensional implementation.
$\mathcal{Z}, \mathcal{L}, \mathcal{X}$	Experience space, metadata space, and marked experience space.
μ_k	Probability distribution over marked memory records.
$\mathcal{A}_k$	Externally grounded reference material with source and version information.
h_k	Additional state required to determine future transition laws.
S_k	Complete sufficient state defined in (1).
S_k^w	Reference state after the external update.
K_{rep}, K_{rec}, K_{cf}	Replay, recombination, and model-relative counterfactual kernels.
Q_k	Proposal distribution induced by the mixture of component kernels.
$\alpha_k, \beta_k, \gamma_k; \rho_k$	Mixture coefficients and exploratory operator share.
$a_k, \mathcal{I}_k$	Absolute candidate-admission rule and admitted index set.
N_k, C_k, A_k	Novelty score, coherence cost, and applicable anchor-disagreement cost.
$T, \omega_{k,j}$	Optional ranking temperature and admitted-candidate weights.
$\hat{Q}_k$	Weighted empirical distribution of admitted learning material.
f_k, J_k	Anchored fitting objective and proximally regularized objective.
λ_G, λ_p	Fitting-anchor and proximal weights.
$\varepsilon_k, \varepsilon_k^{stat}$	Reference-objective tolerance and stationarity-residual tolerance.
η_k	Weight of admitted material in the reference memory-mixture update.
u_k^w, u_k^*	Reference and proposed parameter-memory pairs.
$g_k, L_q, \delta_{q,k}$	Joint acceptance gate, protected loss, and its declared tolerance.
B_k, ϕ_k, σ_k	Finite novelty reference, frozen representation, and positive scale.
P_k, P, π	Cycle kernel, time-homogeneous special case, and invariant measure when one exists.

Appendix C. Analytic Counterexamples and Boundary Cases

These examples are constructed from the definitions to delimit their scope. They are not measurements of an OD implementation.

C.1 Full support does not mean typicality

Let a scalar memory distribution be $\mathcal{N}(0,1)$. Its support is $\mathbb{R}$, so $d(z, \text{supp } \mu) = 0$ for every real z . A generator concentrated at $z = 100$ would therefore have zero support-distance novelty despite being far from the distribution's typical region. Equation (19) avoids this particular degeneracy by using a declared finite reference; it does not solve every difficulty of novelty measurement.

If exact replay has zero distance and exploratory samples have conditional mean distance c , sampling from a mixture with exploratory share ρ already gives mean distance ρc . Multiplying that mean by ρ again gives $\rho^2 c$. Such a quantity could be defined intentionally, but should not be mistaken for an unmodified mixture-average novelty.

C.2 A softmax does not reject an invalid batch

With a single candidate of any finite score r , normalized Gibbs weighting gives $\exp(r/T)/\exp(r/T) = 1$. For several equally scored invalid candidates, it gives uniform weights. No absolute standard enters either calculation. The admission rule in (5), together with the empty-set branch, is therefore distinct from and logically prior to the weighting in (7).

C.3 Soft anchoring can allow anchor degradation

Let the reference parameter be $\theta^w = 0$, the synthetic loss be $(\theta - 10)^2$, and the anchor loss be θ^2 . Set both λ_G and λ_p to one. Then

$$J(\theta) = (\theta - 10)^2 + \theta^2 + \frac{1}{2}\theta^2.$$

The derivative is $5\theta - 20$, so the minimum is $\theta^* = 4$. The anchor loss rises from zero to sixteen even though both penalties are active. A zero-tolerance anchor check would reject the proposal. This example does not contradict the parameter bound; it demonstrates why that bound is not a non-regression theorem.

C.4 One-step bounds do not establish lasting stability

Consider hypothetical accepted scalar changes of size $r > 0$, all in the same direction and without an external reset. Each satisfies a radius bound r , but after n steps the displacement is nr . Likewise, repeated memory mixtures can drive the original distribution's weight toward zero, as described in Section 8.2. Horizon-wide guarantees require additional assumptions, such as a fixed external reference, cumulative constraints, or a suitable contraction or drift condition.

C.5 A stochastic update need not be Markov in the recorded state

Two optimizers can have identical current parameters and memory but different momentum vectors. The next parameter update can then have different conditional laws. Recording only (θ, μ) loses information relevant to prediction. Including the required optimizer state in h resolves this example; it does not prove sufficiency for every other source of hidden history.

C.6 Invariance and continuity do not imply useful learning

For the identity kernel $P(s, \cdot) = \delta_s$, every probability measure is invariant. This can describe an implementation that rejects all changes. It satisfies the invariance equation but provides no evidence of learning.

Conversely, on the compact interval $[-1, 1]$, the deterministic map $F(s) = 0$ for $s < 0$ and $F(s) = 1$ for $s \geq 0$ induces a non-Feller kernel: for the continuous test function $\varphi(s) = s$, $P\varphi = F$ is discontinuous. It nevertheless has an invariant measure, δ_1 . Failure of a sufficient Feller condition therefore does not prove nonexistence; it means that the stated theorem does not establish existence by that route.

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